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OBJ file format

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AI Overview

The Wavefront .obj file format is an open, text-based 3D geometry definition standard originally developed by Wavefront Technologies. Wikipedia +1



Key Characteristics

- **Human-Readable ASCII:** Stored as plain text with space-separated tokens, making it easy to open and inspect in any text editor. scanology +2
- **No Animation:** Supports static 3D geometry only and does not store rigging, bones, or time-based animation sequences. Scratchapixel +1
- **Universal Compatibility:** Supported across major 3D software packages including Blender, Maya, 3ds Max, and Unity. YouTube · Hans IT Academy

Core File Structure & Keywords

An OBJ file reads line-by-line from top to bottom, using primary command prefixes:

- `v` : Geometric vertex coordinates (x, y, z). Paul Bourke +1
- `vt` : Texture coordinate (UV) vertices for mapping 2D images onto 3D surfaces. Wikipedia +1
- `vn` : Vertex normals for smooth or sharp surface shading calculation. Rochester Institute of T... +1
- `f` : Face definitions listing vertex indices (supports triangles, quads, and polygons). Sloyd +1
- `g` / `o` : Group names or object names to organize mesh parts. Paul Bourke

Companion Files

- `[.mtl]` **(Material Template Library):** An external ASCII text file referenced via `mtllib` that defines surface colors, reflectivity, and ambient/diffuse/specular lighting properties using the Phong model. FileFormat.Info +1
- **Textures:** Image files (like PNG or JPG) mapped via the MTL file for surface detailing. scanology +1

OBJ vs. Alternative Formats

- **vs. STL:** Unlike STL—which flattens everything into a single triangulated mesh without color or material data—OBJ preserves separate components, quad geometry, and UV mapping.
- **vs. FBX / glTF:** FBX and glTF support full animation, skeletal rigs, and modern PBR materials, whereas OBJ is strictly limited to static geometry and basic materials. YouTube · 3D Comparison +4

Are you trying to **open, convert, or write code to parse** an OBJ file, and what software or tool are you using?

Ask anything

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Object Files (.obj)
File Structure: Vertex data (v, vt, vn, vp), attributes (cstype, deg, bmat, step), elements (p, l, f, curv, curv2, surf), body…

Show all

Wikipedia
https://en.wikipedia.org › wiki › Wavefront_.obj_file

Wavefront .obj file

OBJ (or .OBJ) is a **geometry definition file format** first developed by Wavefront Technologies for The Advanced Visualizer animation package. Read more

File format | Face elements | Reference materials | Material template library

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These files must have the extension .obj. In this release, the **.obj file format** supports both polygonal objects and free-form objects. Polygonal geometry ... Read more

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Parsing OBJ Files

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Obj File Format

The OBJ file format is a **text file format**, which means you can edit OBJ files in a text editor if you are hard-core. Unfortunately, the original specification ... Read more

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